Love of Variety Model and Simulation

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1 Model

1.1 Consumer's Problem

The consumer's problem is to maximize utility subject to their choice set. The utility function is as follows:

$$U_{ijt} = \beta_i X_{jt} - \gamma_i (\Sigma_{ijt})^2 + \kappa a_{jt} - \alpha p_{jt} + \xi_j + \epsilon_{ijt} \tag{1}$$

$$\Sigma_{ijt} = X_{jt} - \bar{X}_{it}$$

Here, U_{ijt} is the utility of consumer i who chooses product j in period t. Utility depends on utility from characteristics associated with product X_jt, β_i, love of variety γ_i, how different your current choice is from the mean Σ_{ijt}, advertising a_{jt}, prices p_{jt}, and unobserved quality ξ_j. Further, a gumbel shock ε_{ijt} also impacts utility. This is what allows us to construct choice probabilities as follows:

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}} \tag{2}$$

2 Simulation

To understand model behavior, I first do Monte Carlo exercises across different environments. This will be replaced by Nielsen scanner data once access is gained.

2.1 Parameters

Parameter	Value	Description
β_i	{1, 2.5, 4}	Consumer preference for product attributes
γ_i	{1, 2.5, 4}	Consumer love of variety
κ	3	Advertising cost parameter
α_j	0	Price sensitivity
ξ_j	0	Unobserved quality

Table 1: Simulation Parameters

Parameters were informed by the gumbel distribution for scaling. Prices and unobserved quality are set to 0 for simplicity in early phases of testing. Will be added as I get further into the project.

2.2 Simulation Procedure

I have simulated two specifications so far: one in which the mean is fixed as the mean of an ordinal product set and one with a mean informed by "prior preferences" over products from a normal distribution. Below I go into each of these with graphs of choice probabilities in a market with one consumer.

2.3 Set Mean

In the first simulation routine, The product set J is defined as $J = \{1.0, 2.0, 3.0, 4.0, 5.0\}$. Consider this the most simple form available, where $\bar{X}_i t = 2.5$, the mean of J. Thus, $\Sigma_{ijt} = X_{jt} - 2.5$. Therefore, the only thing changing over time is the new epsilon draw each period. This specification also sets $a_{jt} = 0$ to really allow the β_i, γ_i regimes to show their behavior.

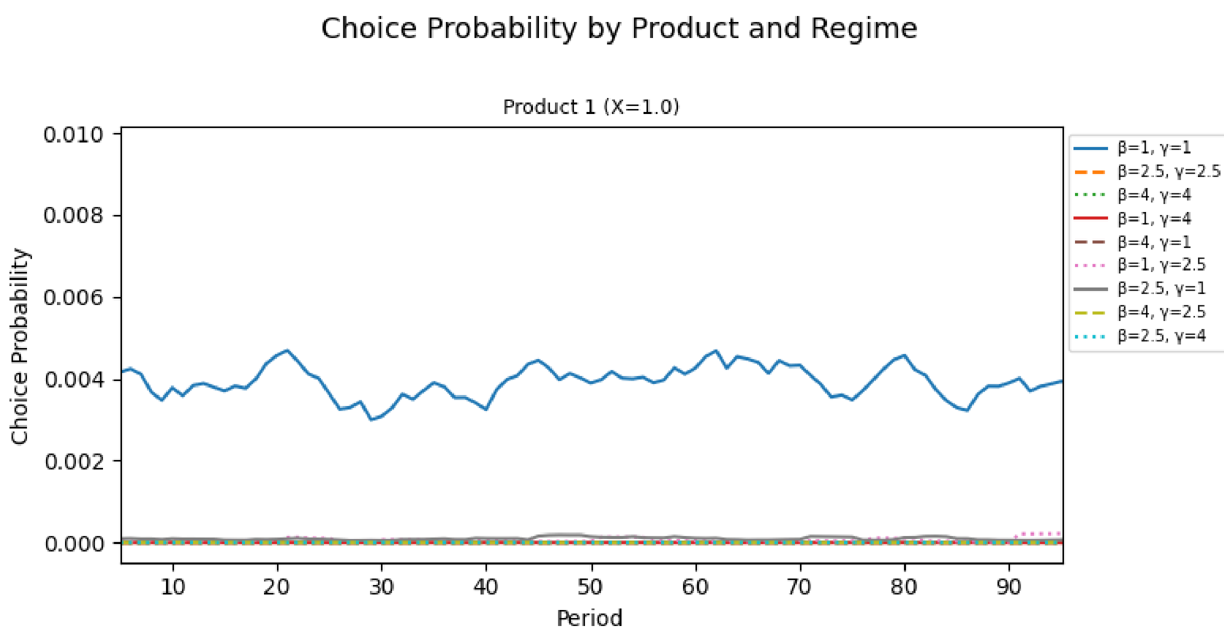


Figure 1: Choice Probability of Product 1 by Regime

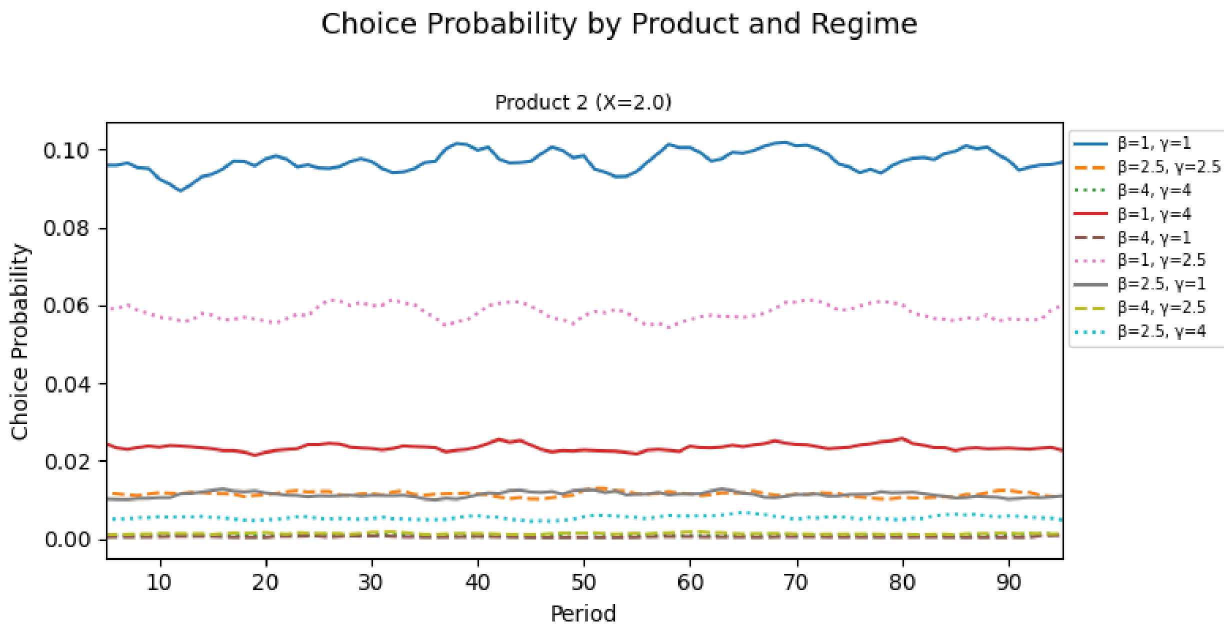


Figure 2: Choice Probability of Product 2 by Regime

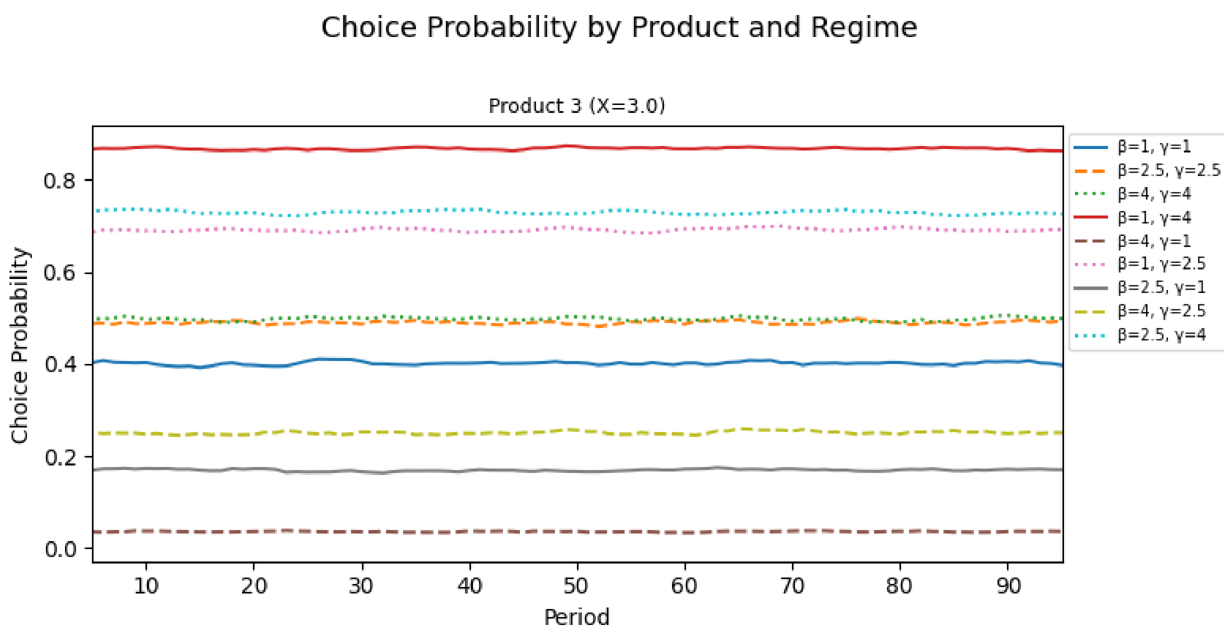


Figure 3: Choice Probability of Product 3 by Regime

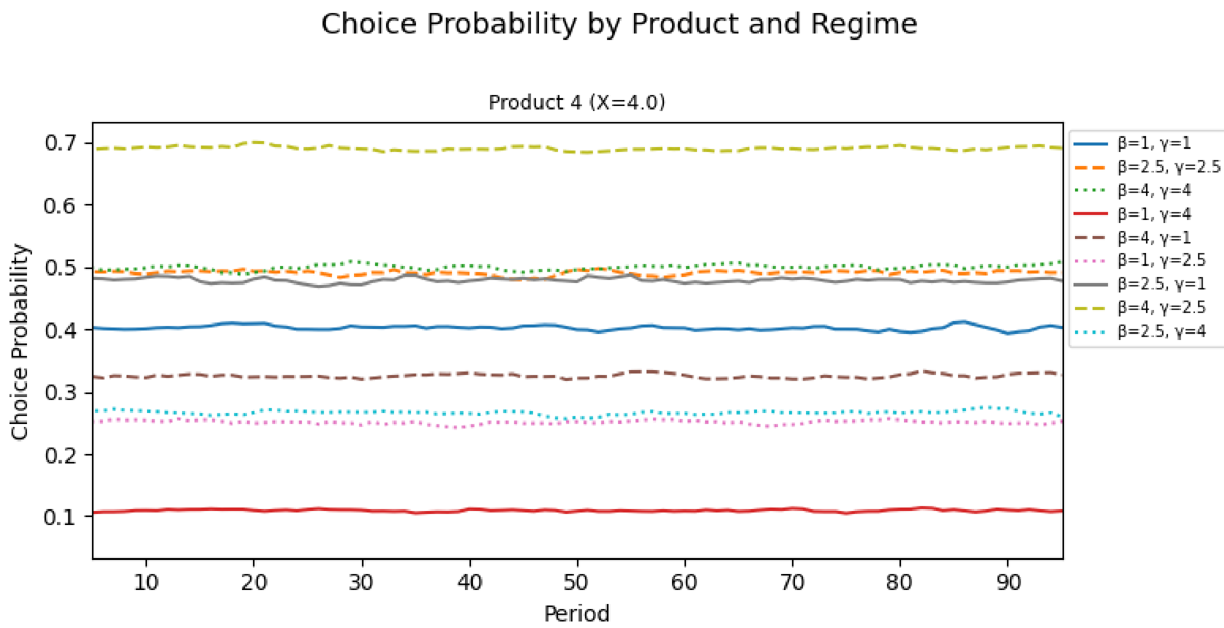


Figure 4: Choice Probability of Product 4 by Regime

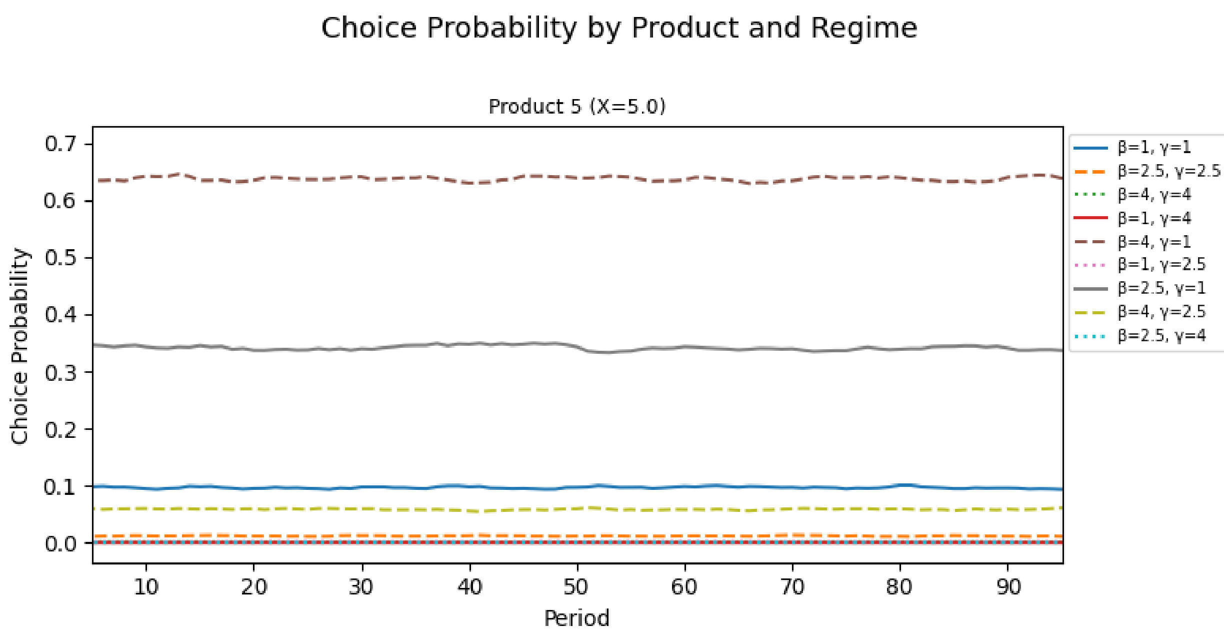
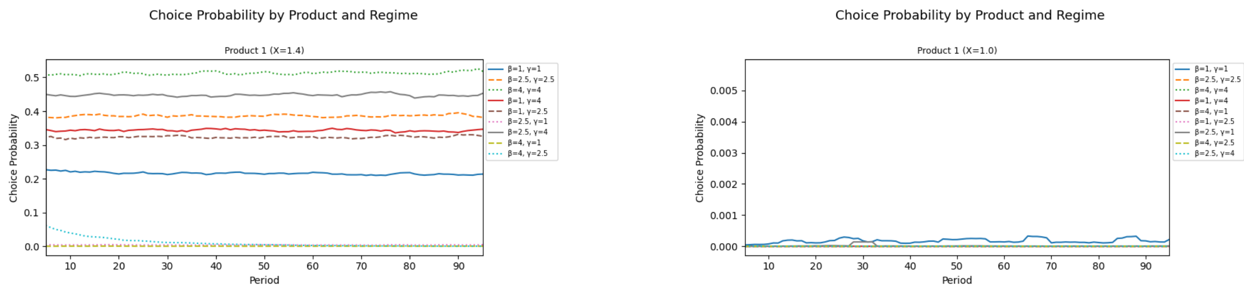


Figure 5: Choice Probability of Product 5 by Regime (Fixed Advertising $a = 0.2$)

In this setup, choice probabilities are practically flat, as is expected. I will pick out a few lines. In product 3, the ($\beta = 1,\ \gamma = 4$) regime has the the highest probability of choosing this product. This is to be expected, as they want to minimize the γ term by getting as close to the mean as possible. For product 4, the ($\beta = 4,\ \gamma = 2.5$) has the highest probability. This is because their utility gain outweighs the penalty enough to choose a higher utility product, though not enough to choose the highest utility product. Product 5, is chosen most by the ($\beta = 4,\ \gamma = 1$) regime for similar reasons as descibed above. The low penalty allows them to "indulge" in a higher utility product.

2.4 Prior Mean

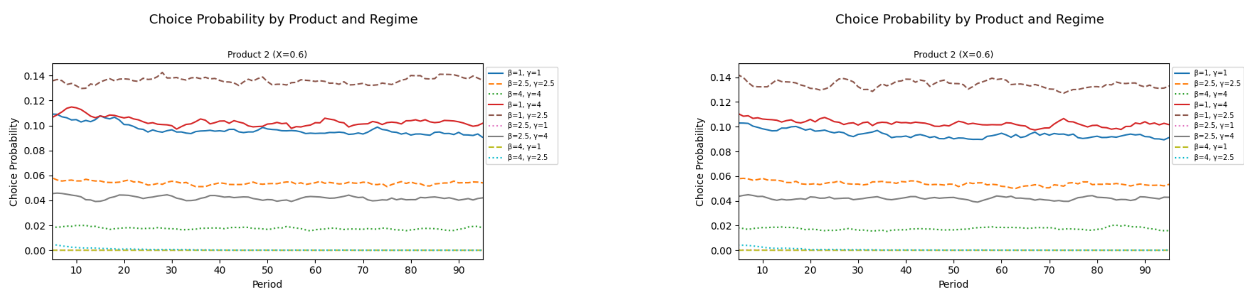
In this routine, I implemented an idea Josh had in a meeting: the consumer has priors over what they like. So, to implement a version of this, I sim five periods before my main simulation of 100 periods to create a mean over X's as they enter the model. Further, I simulated it with $J \sim N(0, 1)$ with 5 products. I did this because in the case from the above section, regimes went to $X = \{4, 5\}$. I wanted to do this to test how I have things specified. Below are choice probabilities for the set advertising case ($\bar{2}.0$) and the evolving advertising case ($\bar{X}_it - X_{jt-1}$).



(a) Choice Probability of Product 1 by Regime (Fixed Advertising $a = 2$)

(b) Choice Probability of Product 1 by Regime (Difference from Mean)

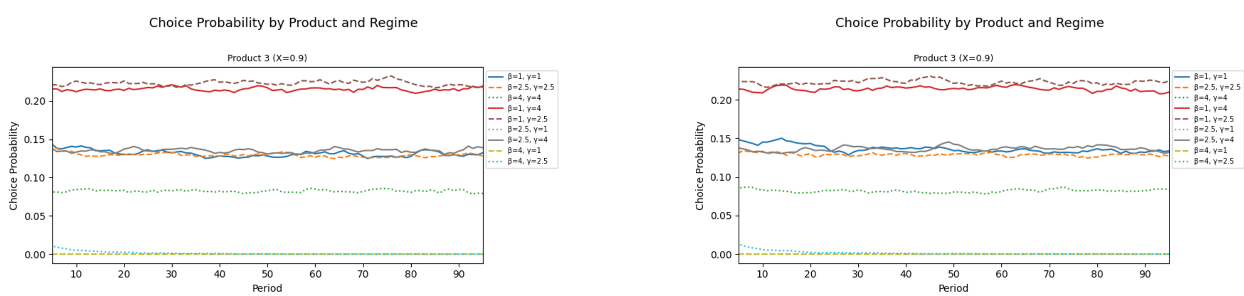
Figure 6: Choice Probability of Product 1 across Regimes



(a) Choice Probability of Product 2 by Regime (Fixed Advertising $a = 2$)

(b) Choice Probability of Product 2 by Regime (Difference from Mean)

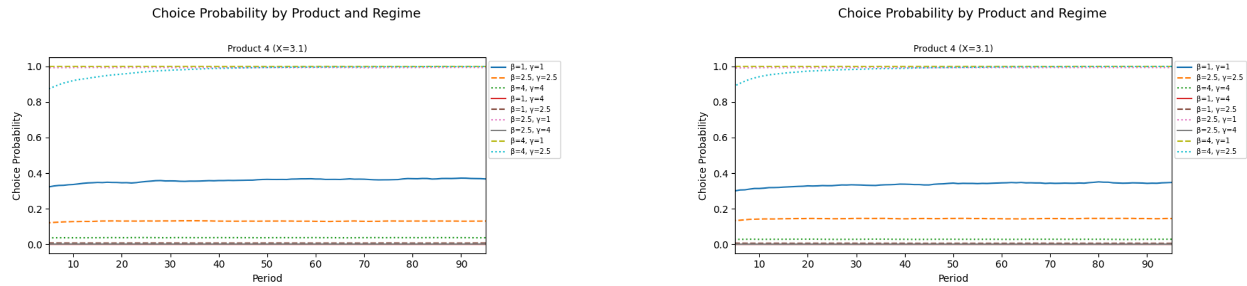
Figure 7: Choice Probability of Product 2 across Regimes



(a) Choice Probability of Product 3 by Regime (Fixed Advertising $a = 2$)

(b) Choice Probability of Product 3 by Regime (Difference from Mean)

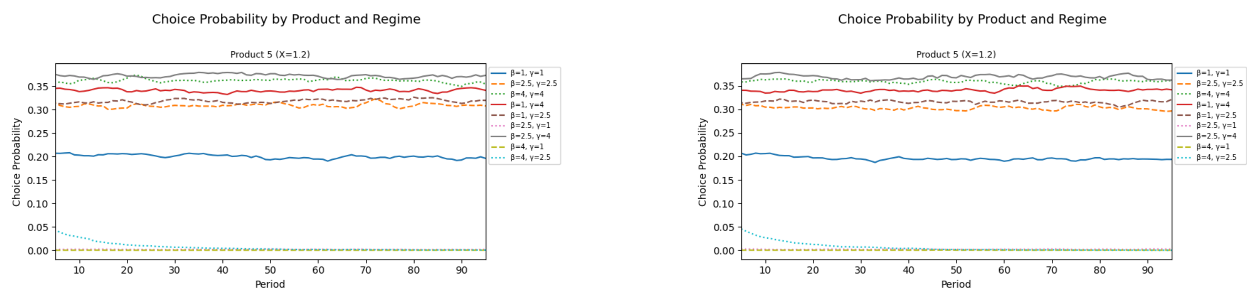
Figure 8: Choice Probability of Product 3 across Regimes



(a) Choice Probability of Product 4 by Regime (Fixed Advertising $a = 2$)

(b) Choice Probability of Product 4 by Regime (Difference from Mean)

Figure 9: Choice Probability of Product 4 across Regimes



(a) Choice Probability of Product 5 by Regime (Fixed Advertising $a = 2$)

(b) Choice Probability of Product 5 Regime (Difference from Mean)

Figure 10: Choice Probability of Product 5 across Regimes

In this specification, we see similar patterns. If a regime is high β and low γ, they choose the highest value product, like product 4. In fixed ads, regimes in the middle, like ($\beta = 1,\ \gamma = 2.5$), will spread choices across 1, 2, 3, 5. This is the behavior I am most interested in: sampling products if things get too "samey". In the specification where ads are based on difference from the mean, product 1 is basically taken out of the model, indicating there needs to be work on this front.

3 Next Steps

Next, I would like to go to 5 consumers, and see how things change. I also want to see how pulling from $\gamma \sim N(0, \sigma_\gamma)$ affects things. Of course, I also need to start really thinking about the firm side and the consumer's value function. I am also working on beginning my literature review, which has inspired some interesting ideas.